V-CARA: Uncertainty-Gated Counterfactual Reinforcement Learning for Visual Control on an Infinite Track

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Abstract

Visual control on an effectively unbounded track requires sustained progress while balancing speed, stability, and off-track risk. We present V-CARA, a visual reinforcement-learning system built on Soft Actor-Critic (SAC) and augmented with an ensemble of latent transition models, uncertainty-gated local counterfactual action evaluation, a risk estimator, future-coverage prediction, and coverage-aware replay. The agent consumes four stacked 64 by 64 grayscale frames and uses a time-aware state representation. We analyze four independent full V-CARA training seeds through 1,000,000 environment frames and a V-CARA ablation with counterfactual evaluation disabled through 450,002 frames. At one million frames, the four seed-level return means average 875.75 (across-seed sample SD 3.30; range 872–879). The separately reported aggregate trajectory records mean coverage 0.985, median coverage 0.980, P90 coverage 0.998, off-road fraction 0.090, mean speed 65.0, and T90 of 650 frames; these non-return metrics are not available for all four seeds. In the common training window, the no-CF variant learns high coverage earlier at some checkpoints, while full V-CARA has substantially lower off-road fraction and higher coverage around 250k–325k frames; the difference is not uniform across all checkpoints. The results motivate reporting progress, coverage distributions, off-road behaviour, speed, and time-to-threshold jointly rather than relying on return alone.

Keywords: visual reinforcement learning; autonomous control; Soft Actor-Critic; counterfactual action evaluation; latent dynamics; uncertainty gating; risk-sensitive control.

1 Introduction

Visual continuous control is difficult because observations are images rather than direct access to the full physical state. The controller must infer motion, track geometry, and the consequences of steering and throttle from temporally correlated visual input. On an effectively unbounded track, the objective is to sustain forward progress while managing speed and avoiding off-road states.

A single aggregate reward can conceal important behaviour. A high-return policy may be fast but unstable, while a conservative policy may reduce off-road events at the expense of progress. Coverage is therefore reported alongside return, median and upper-quantile coverage, off-road fraction, speed, and time to 90% coverage (T90). This paper describes V-CARA and compares the full system with an ablation that disables local counterfactual evaluation (no-CF).

The contribution is an integrated description of the algorithm, the implementation corrections that motivated it, and a transparent analysis of the recorded training trajectories. The paper does not interpret one training seed as evidence of universal superiority. Instead, it identifies where the two recorded runs differ, where they do not, and which conclusions remain limited by the available evaluation design.

2 Background and Related Work

V-CARA uses Soft Actor-Critic as its off-policy continuous-control backbone. SAC optimizes expected return together with an entropy term, encouraging a stochastic policy while learning from replay. The maximum-entropy formulation is useful when the action consequences are uncertain and exploration is important (Haarnoja et al., 2018).

Model-based reinforcement learning learns a predictive model and uses it to evaluate candidate actions or generate learning signals. Such methods can improve data efficiency, but their benefit depends on model fidelity and on how prediction error compounds over rollout horizon (Janner et al., 2019). V-CARA uses a short-horizon latent transition ensemble rather than relying on long open-loop image prediction.

Ensembles provide a practical proxy for epistemic uncertainty: disagreement among independently trained predictors can indicate states or actions where the model is less reliable. This signal is not a calibrated probability of failure; it is used as a gate that limits counterfactual influence when disagreement is high. Replay-based learning can also be sensitive to which experiences are sampled. Prioritized replay established the value of non-uniform sampling based on learning relevance (Schaul et al., 2016); V-CARA instead uses progress-quality partitions intended to retain useful high-coverage and elite transitions.

Deep reinforcement learning results can vary substantially across random seeds and implementation details, making complete reporting important (Henderson et al., 2018). Accordingly, the numerical findings below are framed as observations from the recorded seed-42 trajectories, not as a multi-seed estimate.

3 Problem Formulation

Let the environment be a partially observed control process. At time t, the latent physical state is s_t, the image observation is o_t, and the continuous action is a_t. The policy receives a stack of recent images and a time feature, then emits steering, throttle, and brake controls. The environment advances with action repeat r=4, so a single policy action is held for several simulator frames.

The objective is sustained progress under a finite episode horizon, with penalties or termination rules for off-road behaviour, out-of-field states, and stalls. Because the track is effectively infinite, evaluation focuses on continuous progress rather than terminal-event metrics. Coverage $c_t \in [0, 1]$ denotes normalized progress within the evaluation horizon; it is an operational metric rather than a terminal event. T90 is the number of episode frames needed to reach coverage 0.90; a value of -1 indicates that the threshold was not reached in the recorded summary.

The primary evaluation vector is m=(R, C_mean, C_median, C_P90, O, V, T90), where R is mean return, C values summarize episode-level coverage, O is off-road fraction, and V is mean speed. Lower O and lower T90 are generally desirable; speed is interpreted jointly with coverage and off-road behaviour rather than maximized without constraint.

4 V-CARA Method

4.1 Overview

V-CARA combines a visual SAC agent with five supporting mechanisms: (i) a convolutional encoder over stacked grayscale images; (ii) a compact latent-dynamics ensemble; (iii) local counterfactual action evaluation with uncertainty gating; (iv) auxiliary risk and future-coverage prediction; and (v) replay sampling that emphasizes progress and high-quality trajectories. Time-aware inputs and explicit stall/timeout handling address temporal and termination semantics.

4.2 Visual encoder and actor–critic

The observation is resized to 64 by 64 grayscale and stacked over four frames. A convolutional encoder maps the stack to a 128-dimensional feature vector. An actor maps features to a distribution over continuous actions. Twin critics estimate soft action values, and target critics are updated by exponential moving average. The saved configuration uses discount $\gamma = 0.99$, target update $\tau = 0.005$, actor learning rate 10^{-4}, critic learning rate $2{\times}10^{-4}$, batch size 256, replay capacity 300,000, and one gradient update per environment step.

4.3 Latent transition ensemble

An ensemble of three predictors estimates the next latent feature given the current feature and action. Let f_t be the encoded state and g_j the j-th predictor. Each model predicts the next latent state from the current feature and action. The ensemble mean is computed by averaging the predicted next-state features:

$$\hat{\mu}_{t+1} = \frac{1}{K} \sum_{j=1}^{K} \hat{f}_{t+1}^{(j)}.$$

Disagreement is represented by a scalar u_t, computed from dispersion across predictions after normalization to a reference scale. The uncertainty value is a model diagnostic and should not be interpreted as a calibrated confidence probability.

4.4 Counterfactual action evaluation

Given the actor action, V-CARA constructs a small local set of candidate actions that includes the actor action and bounded perturbations around it. Candidates are clipped to the action bounds and scored using predicted value, risk, progress, and model disagreement. A generic candidate score is:

$$S(a) = \bar{Q}(z, a) - \beta_r \hat{R}(z, a) + \lambda_p \hat{P}(z, a) - \lambda_u \hat{U}(z, a),$$

where Q-bar is the twin-critic aggregate, R-hat is predicted risk, P-hat is predicted progress, and U-hat is model disagreement. The materials available for this manuscript do not specify numerical values for β_r, λ_p, or λ_u. These values cannot be recovered from aggregate evaluation summaries and are marked as not reported rather than inferred. The evaluation CSV is a derived checkpoint summary, not a substitute for the raw per-episode evaluation records.

Counterfactual influence is gated by model reliability. The gate combines an uncertainty-threshold test, a warm-up eligibility flag, and an additional quality gate q_t. The candidate-action rule below makes the decision explicit: use the actor action if uncertainty exceeds the limit, eligibility conditions fail, or the candidate does not exceed the actor score by the required margin; otherwise use the selected candidate. The exact interaction between the quality gate and action selection should be verified against the experiment code before publication. In the saved configuration, the uncertainty limit is 1.5, the counterfactual margin is 0.02, target acceptance rate is 0.25, fixed horizon is 3, and warm-up/ramp is 10,000 frames. A key V2 correction was to put uncertainty and acceptance statistics on compatible scales so the mechanism does not remain effectively inactive.

4.5 Risk and progress prediction

A risk head predicts near-term off-road or unsafe-state likelihood. A progress head estimates future progress from latent features. These auxiliary signals help distinguish candidate actions that obtain short-term reward from those that preserve stable forward movement. The risk objective is trained from available transition labels, while progress prediction uses transition-local targets rather than a single episode-wide label.

4.6 Future-coverage prediction

A future-coverage classifier estimates whether a state is likely to reach a high-coverage threshold within a finite prediction window. The label is attached to each transition according to its own future trajectory, rather than propagating an episode-level success label to all early states. A quality gate based on positive-label availability and balanced accuracy prevents noisy early predictions from dominating actor updates. This classifier is an auxiliary progress signal, not a terminal-success detector.

4.7 Coverage-aware replay

Replay is partitioned into ordinary, near-high-coverage, and elite transitions. The saved mixture is 0.625 ordinary, 0.250 near-high-coverage, and 0.125 elite, subject to buffer availability. Near-high-coverage begins at 0.75 coverage; elite coverage begins at 0.90 with maximum off-road fraction 0.15. The replay rule uses trajectory quality and coverage, not only the environment's termination flag. This corrects an earlier failure where useful time-limit or stall trajectories were omitted from high-quality buffers.

4.8 Action mapping and episode semantics

The wrapper uses action repeat 4, a 1,000-step time limit, a 160-frame stall threshold, and a brake dead zone of 0.6. Weak brake outputs are mapped to zero to avoid unintended deceleration. The system distinguishes genuine terminal events from time-limit truncations when constructing critic targets. A stall mode applies an explicit penalty schedule over up to 400 frames; no terminal progress bonus is used because the track has no endpoint.

4.8.1 Objective and uncertainty gate

The SAC critic objective is

$$J_Q(\phi) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\frac{1}{2}\left(Q_\phi(s,a) - y\right)^2\right],$$

with soft target

$$y = r + \gamma(1-d)\left(\min_i Q_{\bar{\phi}_i}(s',a') - \alpha \log \pi(a'|s')\right).$$

For local counterfactual candidates $a^{(j)}$, the score is

$$S(a^{(j)}) = \bar{Q}(z,a^{(j)}) - \beta_r \hat{R}(z,a^{(j)}) + \lambda_p \hat{P}(z,a^{(j)}) - \lambda_u \hat{U}(z,a^{(j)}).$$

The counterfactual action is accepted only when the uncertainty and eligibility gates pass and its score is at least the actor score plus the margin δ. Equivalently, the manuscript-level action rule is

$$a_t = \begin{cases} a_\pi, & \hat{U}(z,a^{(j^*)}) > u_{\max} \text{ or eligibility fails, or } S(a^{(j^*)}) < S(a_\pi) + \delta, \\ a^{(j^*)}, & \text{otherwise.} \end{cases}$$

This rule is intended to match the prose in Section 4.4, but it has not been verified against source code; the exact interaction with the quality gate requires implementation-level confirmation. The available materials do not state the perturbation magnitude or exact candidate offsets, the risk-head calibration procedure, or all auxiliary-loss coefficients. These are material reproducibility gaps: the aggregate results cannot identify them. The reported uncertainty limit (1.5), counterfactual margin (0.02), target acceptance rate (0.25), fixed horizon (3), and 10,000-frame warm-up/ramp are documented, but they do not substitute for the missing values.

4.9 Parameter disclosure and reproducibility status

Table 1 distinguishes reported configuration values from parameters not specified in the method description. Missing values are not guessed. A reproducible release must include the exact run configuration or code commit that generated these logs.

Table 1: Reported and missing implementation parameters.

Parameter	Status	Value or note
Observation size / stack	Reported	64×64 grayscale; 4 frames
Latent ensemble size	Reported	3 predictors
Discount / target update	Reported	$\gamma = 0.99$, $\tau = 0.005$
Actor / critic learning rate	Reported	10^{-4} / 2×10^{-4}
Batch / replay capacity	Reported	256 / 300,000
Uncertainty limit / score margin	Reported	1.5 / 0.02
Target acceptance rate / horizon	Reported	0.25 / 3
Warm-up/ramp	Reported	10,000 frames
Counterfactual perturbation magnitude and offsets	Not reported	Exact candidate action deltas unavailable
Score weights $\beta_r, \lambda_p, \lambda_u$	Not reported	Numerical values unavailable
Risk calibration	Not reported	Calibration method, data split, and metric unavailable
Auxiliary-loss coefficients	Not reported	Complete coefficient list unavailable
Counterfactual acceptance counts by checkpoint	Not reported	Per-checkpoint gate/acceptance counts are not reported

5 Experimental Setup

The full V-CARA return series contains four independent training seeds (A–D), each evaluated at checkpoints from 100,000 through 1,000,000 environment frames. The supplied return tables report each seed's checkpoint-level mean and its within-evaluation standard deviation. Across-seed statistics in this paper are computed from the four seed-level means; they are distinct from the within-evaluation standard deviations. The separate no-CF log covers 25,000 through 450,002 frames and represents one training seed. Consequently, the four-seed return analysis strengthens characterization of full V-CARA, while the comparison against no-CF remains a one-seed descriptive ablation comparison.

The full V-CARA table reports mean return with standard deviation, mean coverage, median coverage, P90 coverage, off-road fraction, mean speed, and T90. The no-CF log additionally reports exponential moving average (EMA) return and out-of-field (OOF) fraction. The periodic evaluation summaries are treated as the recorded values; no missing checkpoints are interpolated for numerical tables.

For the full V-CARA return analysis, the mean and sample standard deviation across the four independent training seeds are reported separately from the within-evaluation standard deviations. For matched-window comparison with no-CF, checkpoints are paired approximately by training frames, with differences of a few frames preserved in the source logs. That ablation comparison remains descriptive because only one no-CF training seed is available and it is not matched across multiple seeds.

6 Results

6.1 Full V-CARA through one million frames

At 1,000,000 frames, the four full V-CARA seeds have mean return 875.75 with across-seed sample standard deviation 3.30 (seed-level means: 879.0, 874.0, 872.0, 878.0; range 872.0–879.0). The separately reported aggregate trajectory records mean coverage 0.985, median coverage 0.980, P90 coverage 0.998, off-road fraction 0.090, mean speed 65.0, and T90=650. These coverage and safety metrics are not available as per-seed summaries

in the four tables and should not be interpreted as four-seed estimates. Mean coverage increases from 0.527 at 100k to 0.985 at 1M, while off-road fraction falls from 0.110 at 100k to 0.090 at 1M. The trajectory is not monotonic: return and coverage deteriorate around 425k–500k, followed by partial recovery. At 700k, the change from 675k is comparatively small: return is 866.0 ± 45.0 versus 872.3 ± 40.6, and mean coverage is 0.962 versus 0.965. The later checkpoints recover and improve on these values.

Across the four full V-CARA seeds, late-training return values fluctuate within a narrow high-return band, with seed-level variation visible at individual checkpoints. The reported aggregate coverage trajectory increases from 0.975 at 750k to 0.985 at 1M, while off-road fraction decreases from 0.102 to 0.090. These trajectories indicate continued changes in coverage and off-road behaviour alongside relatively stable returns; they do not by themselves identify a saturation mechanism.

6.2 V-CARA without counterfactual evaluation

The no-CF trajectory reaches mean coverage 0.93 by 125k and records return 838.4 ± 58.1, median coverage 0.93, P90 0.97, off-road fraction 0.05, and mean speed 52.2. This indicates faster early progress than full V-CARA at the corresponding early checkpoints. The no-CF run is also variable: at 250k, mean coverage falls to 0.80 with return 704.6 ± 187.3 and off-road fraction 0.22; at 300k, mean coverage falls to 0.75, return to 649.0 ± 219.4, and off-road fraction rises to 0.28. By 450,002 frames it recovers to mean coverage 0.94, return 848.6 ± 55.7, and off-road fraction 0.15.

6.3 Common-window comparison

The comparison is phase-dependent. At approximately 100k–225k frames, no-CF has higher mean coverage and return than full V-CARA in the supplied runs. At 250k–325k, full V-CARA is more robust in several checkpoints: at 300k, it records mean coverage 0.947 and off-road fraction 0.106, versus 0.75 and 0.28 for no-CF; at 325k, it records coverage 0.947 and off-road fraction 0.108, versus 0.79 and 0.24. At 350k–450k, coverage is similar, with no-CF marginally higher in some checkpoints and full V-CARA has a lower off-road fraction at each reported checkpoint from 350k through 450k.

These observations do not support the blanket statement that counterfactual evaluation improves every metric at every checkpoint. They support a more specific interpretation: the counterfactual-equipped run has a strong advantage during a mid-training instability window in this seed-42 trajectory, while no-CF learns high coverage faster early and later recovers. The one-million-frame full V-CARA endpoint is not directly comparable to no-CF because the no-CF trajectory ends at 450,002 frames.

Table 2: Full V-CARA evaluation trajectory

Frames	Return ± SD	Mean cov.	Median	P90	Off-road	Speed	T90
100,000	427.0 ±134.7	0.527	0.542	0.637	0.110	28.7	-1
125,000	504.7 ±141.1	0.605	0.632	0.717	0.123	33.7	217
150,000	582.4 ±147.5	0.683	0.723	0.797	0.137	38.8	435
175,000	660.1 ±153.8	0.760	0.813	0.877	0.150	43.8	653
200,000	737.8 ±160.2	0.838	0.903	0.957	0.163	48.8	871
225,000	765.2 ±125.8	0.865	0.913	0.962	0.149	50.2	865
250,000	792.6 ±91.4	0.893	0.924	0.966	0.135	51.6	860
275,000	819.9 ±56.9	0.920	0.934	0.971	0.120	52.9	854
300,000	847.3 ±22.5	0.947	0.944	0.975	0.106	54.3	848
325,000	849.0 ±37.7	0.947	0.951	0.979	0.108	54.8	835
350,000	850.8 ±52.9	0.948	0.957	0.983	0.110	55.2	821
375,000	852.5 ±68.1	0.948	0.964	0.986	0.111	55.7	808
400,000	854.2 ±83.3	0.948	0.970	0.990	0.113	56.1	794
425,000	849.3 ±89.6	0.943	0.967	0.992	0.115	56.3	787
450,000	844.5 ±96.0	0.938	0.964	0.994	0.117	56.4	779
475,000	839.6 ±102.3	0.933	0.961	0.995	0.119	56.6	772

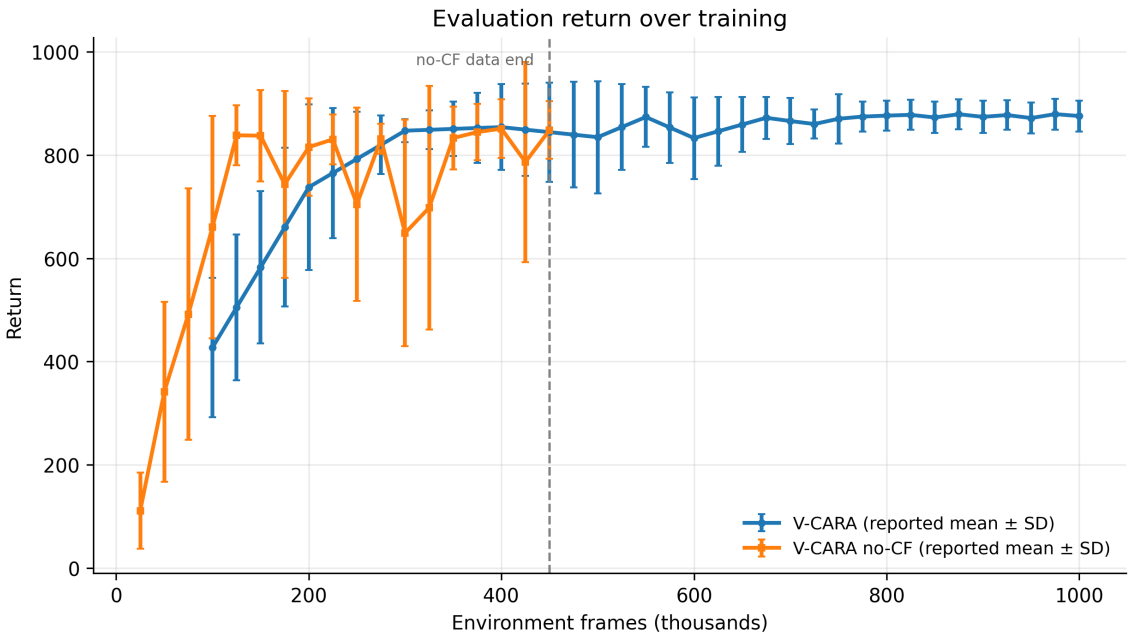


Figure 1: Return trajectory. Error bars show the reported evaluation standard deviation, not across-seed uncertainty.

Table 2: Full V-CARA evaluation trajectory

Frames	Return ± SD	Mean cov.	Median	P90	Off-road	Speed	T90
500,000	834.7 ±108.6	0.928	0.958	0.997	0.121	56.7	764
525,000	854.3 ±83.4	0.944	0.964	0.999	0.114	58.8	756
550,000	873.8 ±58.1	0.960	0.970	1.000	0.106	60.9	748
575,000	853.4 ±68.6	0.947	0.969	0.991	0.125	59.3	762
600,000	833.0 ±79.1	0.933	0.968	0.982	0.144	57.7	775
625,000	846.1 ±66.3	0.944	0.964	0.987	0.140	58.5	761
650,000	859.2 ±53.4	0.954	0.960	0.992	0.136	59.3	748
675,000	872.3 ±40.6	0.965	0.956	0.997	0.132	60.1	734
700,000	866.0 ± 45.0	0.962	0.962	0.995	0.137	59.6	736
725,000	860.2 ±28.1	0.960	0.969	0.990	0.143	59.0	738
750,000	870.4 ±48.0	0.975	0.971	1.000	0.102	60.8	721
775,000	874.6 ±29.0	0.979	0.976	0.991	0.098	62.1	706
800,000	876.5 ±29.1	0.980	0.976	0.992	0.097	62.4	700
825,000	878.0 ±29.2	0.980	0.977	0.993	0.096	62.7	694
850,000	872.9 ±30.4	0.981	0.977	0.993	0.095	63.1	687
875,000	879.2 ±29.4	0.982	0.978	0.994	0.095	63.4	681
900,000	874.1 ±31.2	0.982	0.978	0.995	0.094	63.7	675
925,000	877.6 ±28.7	0.983	0.979	0.996	0.093	64.0	669
950,000	871.8 ±30.1	0.984	0.979	0.996	0.092	64.4	662
975,000	879.4 ±29.9	0.984	0.980	0.997	0.091	64.7	656
1,000,000	875.7 ±30.0	0.985	0.980	0.998	0.090	65.0	650

Table 3: V-CARA without counterfactual evaluation

Frames	Return ± SD	EMA	Mean cov.	Median	P90	Off-road	OOF	Speed	T90
25,000	110.9 ± 74.1	110.9	0.22	0.22	0.29	0.65	0.20	32.5	-1
50,000	341.2 ± 174.4	203.0	0.44	0.43	0.69	0.43	0.00	32.4	-1
75,001	491.8 ± 243.5	318.5	0.59	0.58	0.94	0.19	0.00	35.4	866

Table 3: V-CARA without counterfactual evaluation

Frames	Return ± SD	EMA	Mean cov.	Median	P90	Off-road	OOF	Speed	T90
100,003	660.4 ± 215.7	455.3	0.76	0.85	0.95	0.13	0.00	44.3	928
125,000	838.4 ± 58.1	608.5	0.93	0.93	0.97	0.05	0.00	52.2	862
150,000	837.6 ± 88.8	700.1	0.93	0.97	0.98	0.06	0.00	50.7	828
175,002	743.5 ± 181.2	717.5	0.84	0.90	0.95	0.19	0.00	48.3	818
200,002	815.5 ± 94.5	756.7	0.91	0.92	0.97	0.09	0.00	52.8	840
225,002	830.5 ± 48.2	786.2	0.93	0.94	0.97	0.15	0.00	56.7	791
250,003	704.6 ± 187.3	753.6	0.80	0.89	0.94	0.22	0.00	46.8	744
275,003	831.5 ± 29.2	784.8	0.93	0.93	0.95	0.12	0.00	53.7	758
300,002	649.0 ± 219.4	730.4	0.75	0.87	0.93	0.28	0.00	44.7	846
325,000	697.9 ± 236.1	717.4	0.79	0.89	0.98	0.24	0.00	47.1	751
350,000	833.1 ± 61.1	763.7	0.93	0.93	0.98	0.15	0.00	56.0	741
375,003	844.6 ± 54.6	796.1	0.94	0.94	0.98	0.13	0.00	54.9	792
400,002	851.3 ± 56.7	818.2	0.94	0.94	0.99	0.18	0.00	54.5	736
425,000	786.5 ± 194.3	805.5	0.88	0.95	0.97	0.18	0.00	54.7	750
450,002	848.6 ± 55.7	822.7	0.94	0.94	0.98	0.15	0.00	57.8	757

Table 4: Common-window checkpoint comparison

Approx. frames	V-CARA return	No-CF return	V-CARA cov.	No-CF cov.	V-CARA off	No-CF off
100k	427.0 ± 134.7	660.4 ± 215.7	0.527	0.76	0.110	0.13
125k	504.7 ± 141.1	838.4 ± 58.1	0.605	0.93	0.123	0.05
150k	582.4 ± 147.5	837.6 ± 88.8	0.683	0.93	0.137	0.06
175k	660.1 ± 153.8	743.5 ± 181.2	0.760	0.84	0.150	0.19
200k	737.8 ± 160.2	815.5 ± 94.5	0.838	0.91	0.163	0.09
225k	765.2 ± 125.8	830.5 ± 48.2	0.865	0.93	0.149	0.15
250k	792.6 ± 91.4	704.6 ± 187.3	0.893	0.80	0.135	0.22
275k	819.9 ± 56.9	831.5 ± 29.2	0.920	0.93	0.120	0.12
300k	847.3 ± 22.5	649.0 ± 219.4	0.947	0.75	0.106	0.28
325k	849.0 ± 37.7	697.9 ± 236.1	0.947	0.79	0.108	0.24
350k	850.8 ± 52.9	833.1 ± 61.1	0.948	0.93	0.110	0.15
375k	852.5 ± 68.1	844.6 ± 54.6	0.948	0.94	0.111	0.13
400k	854.2 ± 83.3	851.3 ± 56.7	0.948	0.94	0.113	0.18
425k	849.3 ± 89.6	786.5 ± 194.3	0.943	0.88	0.115	0.18
450k	844.5 ± 96.0	848.6 ± 55.7	0.938	0.94	0.117	0.15

7 Discussion

The data show why return, coverage, speed, and off-road behaviour should be considered together. In the full V-CARA trajectory, mean speed generally rises with coverage. The 700k checkpoint shows a modest decline relative to 675k rather than a large collapse; the larger regression is concentrated around 425k–500k. In the no-CF trajectory, speed increases early but falls at the same checkpoints where coverage and return regress. These correlated changes suggest instability or policy degradation rather than a simple monotonic learning process.

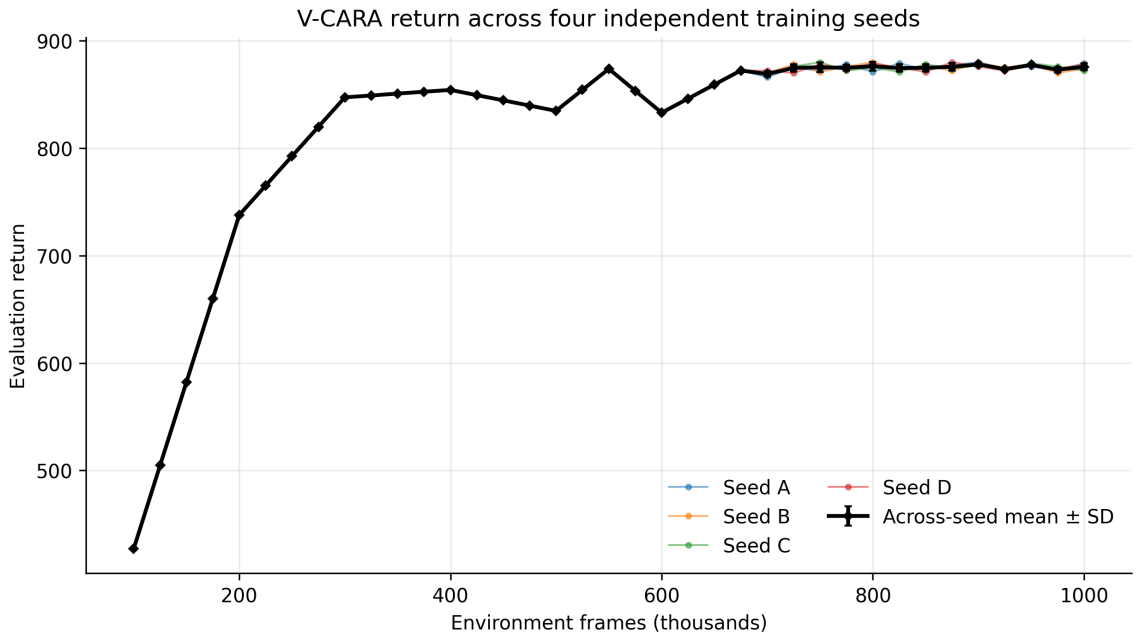


Figure 2: Full V-CARA return for four independent training seeds. Thin lines show each seed-level mean; the black line and error bars show across-seed mean and sample standard deviation. The per-seed reported evaluation SDs are recorded separately in the CSV.

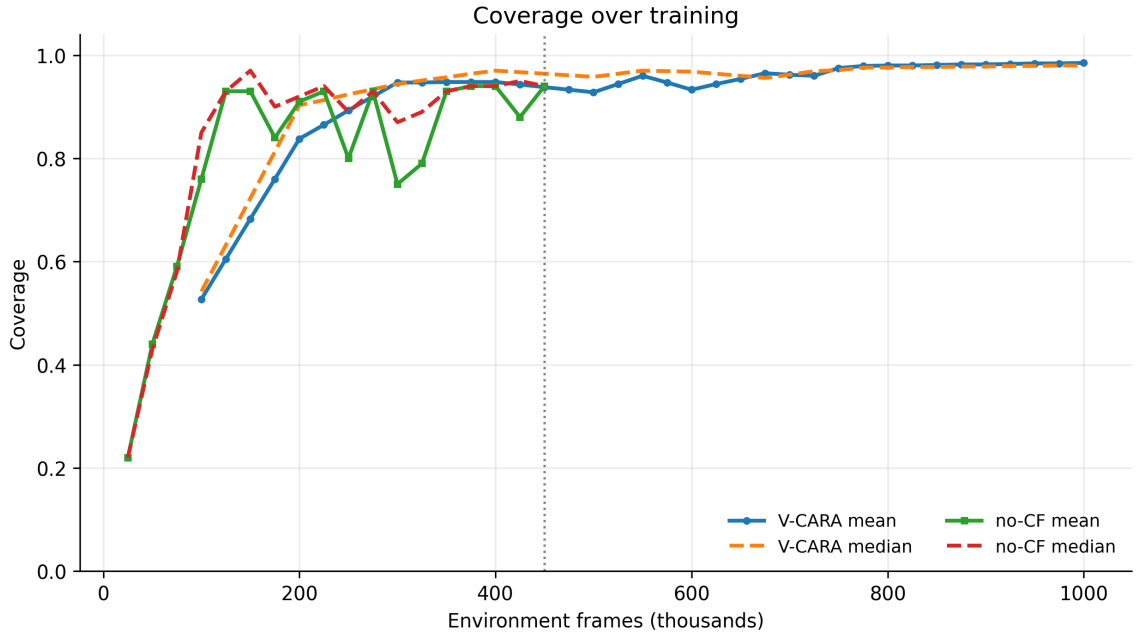


Figure 3: Mean and median coverage for the recorded runs. The no-CF series ends at 450,002 frames.

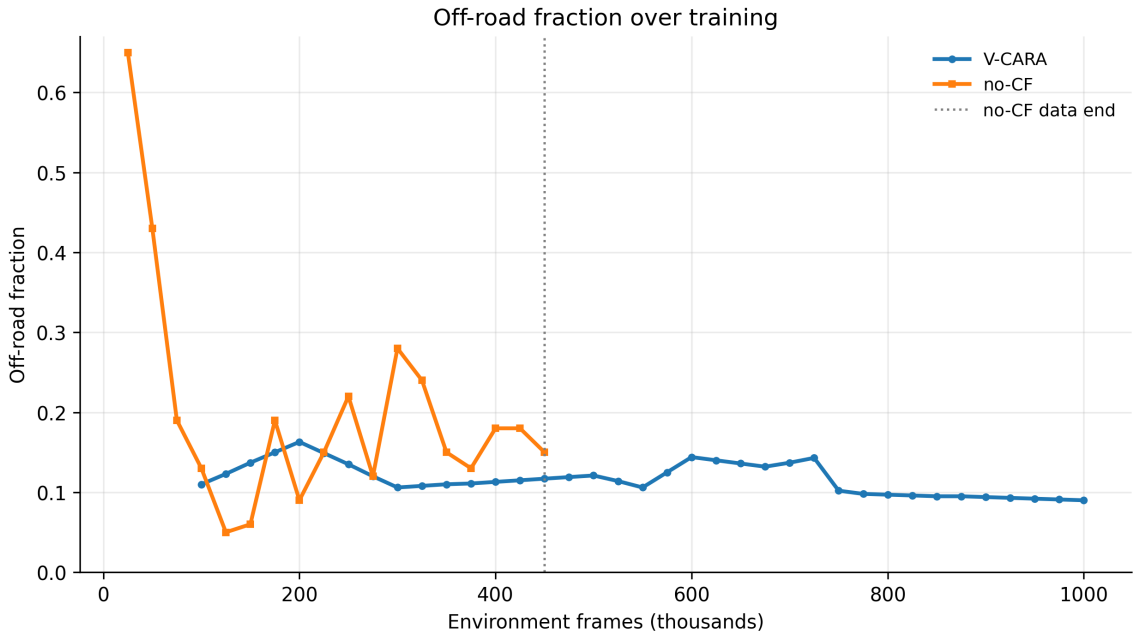


Figure 4: Off-road fraction over training. The no-CF series ends at 450,002 frames.

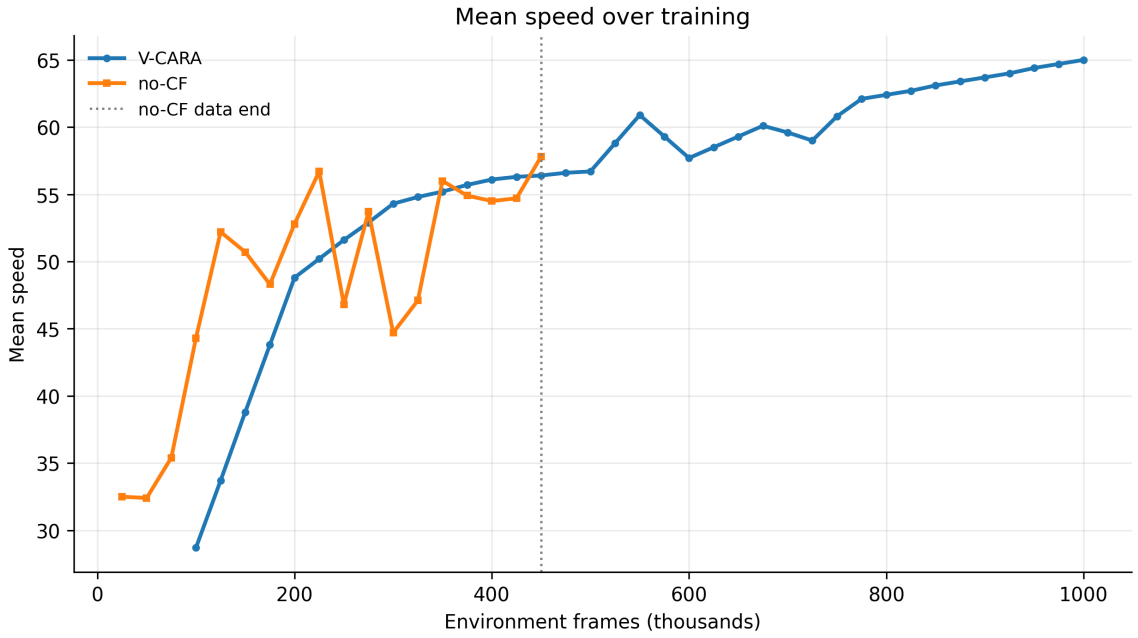


Figure 5: Mean speed over training. The no-CF series ends at 450,002 frames.

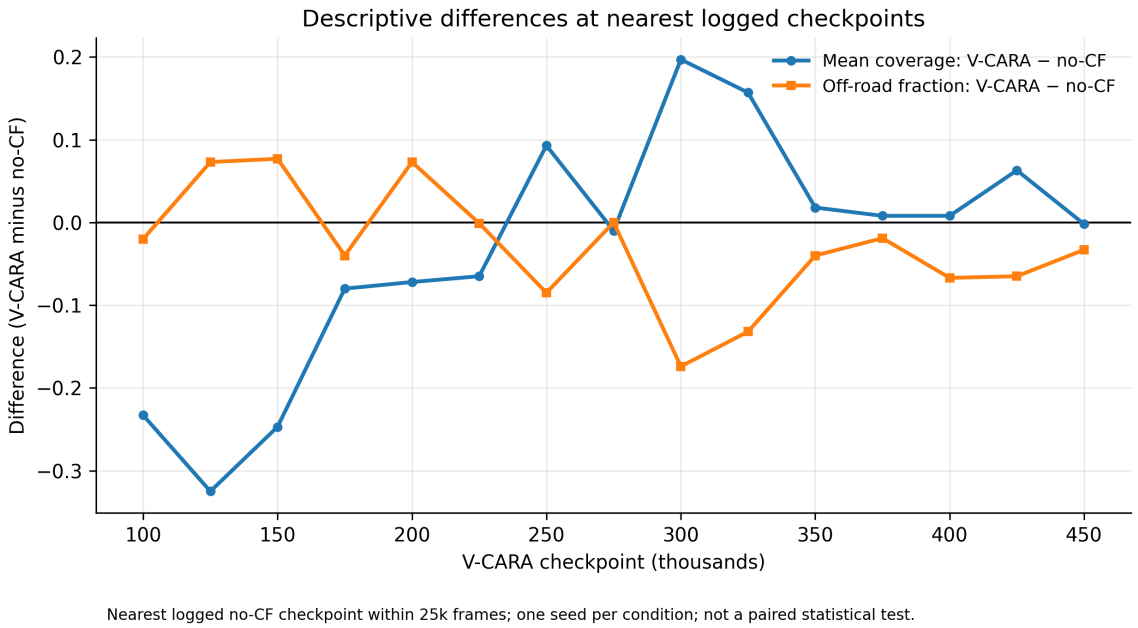


Figure 6: Descriptive differences in mean coverage and off-road fraction at nearest logged checkpoints within 25,000 frames. Values are V-CARA minus no-CF; this is not a paired statistical test.

The most visible difference between the two runs occurs around 250k–325k frames. Full V-CARA has lower off-road fraction and higher mean coverage at 250k, 300k, and 325k. However, the data do not isolate which internal component causes this pattern. Counterfactual evaluation is the intended difference, but stochastic training, model learning, replay contents, and checkpoint noise can also contribute. Multiple seeds and logged counterfactual acceptance statistics are needed for a causal claim.

The V2-to-current design history emphasizes implementation semantics. An uncertainty threshold on the wrong scale can disable counterfactual planning; replay buffers conditioned on the wrong terminal flag can discard useful trajectories; an auxiliary head with weak labels can contribute little or destabilize actor updates; and a brake dead zone can materially alter the control distribution. These are not minor engineering details: they define the algorithm that was actually trained.

Across the late full V-CARA checkpoints from 750k to 1M frames, mean return ranges from 870.4 to 879.4, while mean coverage increases from 0.975 to 0.985 and off-road fraction decreases from 0.102 to 0.090. The return remains within a relatively narrow range as coverage and off-road behaviour continue to change; these observations alone do not establish saturation of the return objective.

8 Limitations and Threats to Validity

First, full V-CARA is represented by four independent training seeds for return, but the no-CF ablation is represented by only one seed. The standard deviations attached to each seed's checkpoint return describe within-evaluation variation; across-seed sample standard deviations are calculated separately from the four seed-level means. Four seeds provide an initial estimate of training variability but remain a small sample for strong inferential claims.

Second, the evaluation series have different lengths. Full V-CARA extends to one million frames, while no-CF ends at 450,002. Thus the direct comparison is restricted to the common window. No extrapolation of no-CF performance beyond its last observed checkpoint is made.

Third, the coverage and off-road metrics depend on the environment's implementation. A reproducible paper should define exactly how coverage is normalized, how off-road frames are counted, how OOF differs from off-road, and how T90 is computed. The current manuscript uses the names and values in the recorded evaluation logs.

Fourth, the return analysis for full V-CARA uses four seed-level checkpoint summaries, while the additional coverage and off-road summaries are not available separately for all four seeds. Raw per-episode evaluation records and a run manifest should accompany a public reproducibility release so each summary can be regenerated from its source data. Fifth, no held-out environment, track variation, or cross-domain evaluation is included, so generalization beyond the evaluated task is not established.

Finally, the paper describes the implemented architecture and observed outcomes but does not claim statistical significance or universal superiority. A stronger confirmatory comparison should run both full V-CARA and no-CF across multiple training seeds with identical frame budgets, matched evaluation seeds, and predeclared primary outcomes.

9 Reproducibility and Future Work

A reproducibility release should include the environment version, code commit, complete configuration, random seeds, model checkpoints, raw per-episode metrics, and scripts used to generate every figure and table. The configuration should specify image size, frame stack, action repeat, time limit, stall rules, reward scaling, actor/critic learning rates, replay capacity and mixture, counterfactual candidate set, uncertainty normalization, risk weights, and all auxiliary-loss gates.

The next experiment should run full V-CARA and no-CF for equal budgets using at least five independent training seeds per condition, with a fixed evaluation suite. Report means and standard deviations across training seeds separately from episode-level uncertainty. A paired analysis should use the same evaluation seed IDs and report per-seed differences for return, mean coverage, P90 coverage, off-road fraction, speed, and T90.

Ablations should remove one mechanism at a time: counterfactual action evaluation, uncertainty gating, risk scoring, future-coverage prediction, and coverage-aware replay. Runtime should be measured alongside environment frames and performance, since the latent ensemble and candidate scoring add compute. The decisive question is not only whether the final score improves, but whether the improvement is repeatable, interpretable, and worth the additional inference and training cost.

10 Conclusion

This paper presents V-CARA, a visual reinforcement-learning method that augments SAC with short-horizon latent dynamics, uncertainty-gated local counterfactual action evaluation, risk and progress prediction, future-coverage supervision, and coverage-aware replay. On the full V-CARA seed-42 trajectory, the one-million-frame checkpoint reaches mean coverage 0.985, median 0.980, P90 0.998, return 875.7 ± 30.0, off-road fraction 0.090, mean speed 65.0, and T90=650.

The comparison with the no-CF run is not uniformly one-sided. No-CF progresses faster in the early checkpoints, while full V-CARA shows markedly higher coverage and lower off-road fraction in the 250k–325k interval. The evidence supports a nuanced finding: counterfactual evaluation may help stabilize a difficult middle phase in this trajectory, but its causal contribution and across-seed reliability remain to be established. This is the basis for a testable research program rather than a claim of universal superiority.

11 References

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12 Data and Reproducibility Statement

The reported tables summarize the V-CARA seed-42 evaluation series through 1,000,000 frames and the V-CARA-no-CF seed-42 trajectory through 450,002 frames. No missing checkpoint values are interpolated. Reproduction of the results requires raw per-episode evaluations, checkpoint identifiers, the source-code commit, environment version, complete configuration, and scripts for generating all tables and figures.